Teardown Benchmark Study

Powertrain and driveline - transmission, propeller shaft and rear axle
Subject vehicle against a comparable-tonnage benchmark

Document number	CVOEM/VAVE/TD-BM/2024/012
Revision	Rev 1.0
Issue date	June 2024
Prepared by	VAVE Benchmarking Cell, Powertrain Installation
Reviewed by	Chief Engineer - Powertrain
Scope	Transmission, propeller shaft, rear axle, mountings
Vehicle class	HCV rigid, 28 t GVW
Distribution	VAVE Committee; Powertrain; Cost Engineering

Summary

The transmission, propeller shaft and rear axle of a comparable-tonnage benchmark were compared against the subject vehicle. The benchmark uses a nine-speed gearbox with a ratio spread of about 11.5:1 against the subject vehicle's six-speed at about 6.8:1. A closer-ratio top-gear cluster is flagged as a fuel-economy study candidate, pending a dedicated drive-cycle trial.

Contents

1	Scope and method	2
2	Driveline layout	4
3	Powertrain and driveline	6
4	Opportunities and conclusions	8

SYNTHETIC
ILLUSTRATIVE - NOT A CVOEM RECORD

1 Scope and method

This study covers the transmission, propeller shaft assembly, rear axle carrier and the associated mountings. Engine internals are excluded. Both vehicles were stripped to component level and the transmissions were opened and the gear sets measured.

1.1 Ratio measurement

Gear ratios were derived from tooth counts measured on the opened transmissions rather than from published data, so that the comparison is of the units actually fitted. Overall driveline ratios combine the measured gearbox ratio with the measured final drive ratio.

1.2 Why ratio spread matters for VAVE

Ratio spread is the ratio between the lowest and highest gear. A wider spread allows a lower first gear for start-ability and a higher top gear for cruise engine speed, without compromising either. A narrower spread forces a compromise, usually taken as a higher cruise engine speed and therefore higher fuel consumption at highway duty.

SYNTHETIC
ILLUSTRATIVE - NOT A CVOEM RECORD

1.3 Measured masses

Component	Subject (kg)	Benchmark C (kg)	Difference
Transmission, dry	286.4	312.8	+26.4
Clutch assembly	42.1	44.6	+2.5
Propeller shaft, front section	38.6	34.2	-4.4
Propeller shaft, rear section	31.2	28.8	-2.4
Centre bearing and mount	8.4	7.1	-1.3
Rear axle carrier	412.6	398.4	-14.2
Transmission mounting	18.9	16.2	-2.7
Total	838.2	842.1	+3.9

The benchmark transmission is 26.4 kg heavier, which is consistent with three additional gear pairs and a longer case. Every other component in the group is lighter on the benchmark, so the net difference is 3.9 kg.

Finding 1.1. A wider ratio spread is not free: it costs 26.4 kg and additional case length. The benchmark recovers most of that mass elsewhere in the driveline, which is the more interesting observation.

SYNTHETIC
ILLUSTRATIVE - NOT A CVOEM RECORD

2 Driveline layout

2.1 Propeller shaft arrangement

Both vehicles use a two-section propeller shaft with a centre bearing. The benchmark uses a 100 mm tube against 110 mm on the subject vehicle, with a thinner wall and a higher-grade steel, giving a lower mass at a comparable critical speed.

Attribute	Subject	Benchmark C
Tube outside diameter (mm)	110	100
Wall thickness (mm)	3.5	4.0
Tube material	IS 3601 Grade 1	Micro-alloyed, higher yield
Front section mass (kg)	38.6	34.2
Calculated critical speed (rpm)	6,840	6,710
Operating maximum (rpm)	3,200	3,150

2.2 Centre bearing

The benchmark centre bearing mount is a pressed steel bracket with a bonded rubber isolator of 7.1 kg. The subject vehicle uses a fabricated bracket of 8.4 kg with a separate isolator and clamp plate. The bonded assembly removes two part numbers and an assembly operation.

Finding 2.1. The bonded centre bearing mount saves 1.3 kg, two part numbers and one assembly operation. It is a running change subject to a durability confirmation on the bonded interface.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD

2.3 Rear axle carrier

The benchmark carrier is 14.2 kg lighter at the same rating. The difference is in the housing wall section away from the bearing bosses, which is 9 mm on the benchmark against 11 mm on the subject vehicle. Bearing boss sections are comparable.

This is the same pattern seen in the frame and the axle beam elsewhere in this benchmarking programme: material distributed uniformly for manufacturing simplicity rather than according to the load path.

Section	Subject (mm)	Benchmark C (mm)
Pinion bearing boss	18	18
Differential bearing boss	16	16
Housing wall, upper	11	9
Housing wall, lower	11	9
Cover flange	14	12

Finding 2.2. A reduced housing wall away from the bearing bosses is worth 14.2 kg. It is a platform change requiring a casting tooling modification and a full axle durability validation.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD

3 Powertrain and driveline

3.1 Transmission ratio comparison

The benchmark uses a nine-speed gearbox with an indicative ratio spread of about 11.5:1, against the subject vehicle's six-speed at about 6.8:1. A closer-ratio top-gear cluster is flagged as a fuel-economy study candidate pending a dedicated drive-cycle trial.

Gear	Subject ratio	Benchmark C ratio
1st	6.80	11.50
2nd	4.02	8.62
3rd	2.44	6.44
4th	1.54	4.82
5th	1.00	3.60
6th	0.78	2.70
7th	-	2.02
8th	-	1.34
9th	-	1.00
Spread	6.8:1	11.5:1

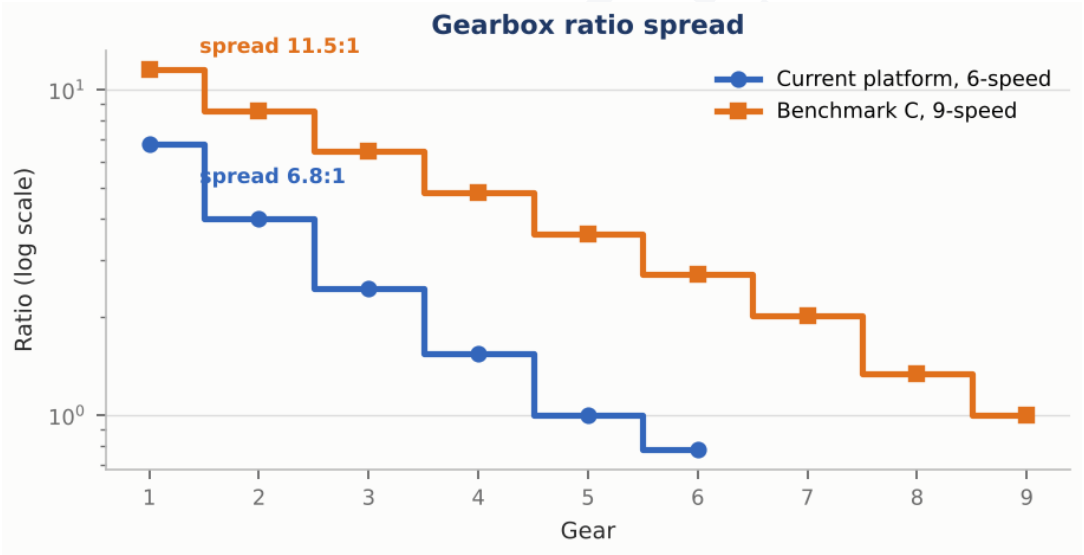


Figure 1 - Gearbox ratio spread. The benchmark's nine-speed layout allows a lower first gear and a closer top-gear cluster within one transmission.

3.2 Consequence at cruise

The subject vehicle's top gear of 0.78 with a 4.1 final drive gives an engine speed of about 1,520 rpm at 80 km/h. The benchmark's 1.00 top gear with a 3.08 final drive gives about 1,310 rpm at the same road speed. On the engine's measured fuel map, that difference corresponds to an estimated 3 to 4 per cent improvement in highway fuel consumption.

Finding 3.1. The fuel benefit arises from the combination of the top gear and the final drive, not from the gear count alone. A proposal that changes only the final drive would reduce start-ability without the wider spread to recover it, and should not be assessed as equivalent.

3.3 What a nine-speed costs

Element	Subject	Benchmark C	Difference
Transmission mass (kg)	286.4	312.8	+26.4
Case length (mm)	712	804	+92
Estimated cost (INR)	184,000	241,000	+57,000
Gear pairs	7	10	+3

The 92 mm additional case length is a packaging constraint on this chassis: the transmission tunnel and the first crossmember would both require modification. This is why the ratio finding is recorded as a study rather than an opportunity.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD

4 Opportunities and conclusions

Ref	Opportunity	Mass (kg)	Cost (INR)	Class
TD-12-01	Bonded centre bearing mount	1.3	210	Running change
TD-12-02	100 mm propeller shaft tube, higher grade	6.8	480	Platform change
TD-12-03	Reduced axle housing wall	14.2	1,840	Platform change
TD-12-04	Closer-ratio top-gear cluster	0.0	-	Study

- The net driveline mass difference is only 3.9 kg, but it conceals a 26.4 kg transmission penalty offset by savings in every other component.
- The ratio spread finding is the most valuable in the study and the least deliverable: 92 mm of additional case length is a chassis packaging change.
- A closer-ratio top-gear cluster within the existing six-speed case is the practical route to part of the fuel benefit and is recorded as TD-12-04.
- Finding 3.1 should be recorded in the idea register so that final-drive-only proposals are not assessed as equivalent to the ratio change.

Recommendation. Release TD-12-01. Open a drive-cycle trial for TD-12-04 before committing to a ratio study, since the fuel benefit estimate rests on a map calculation rather than a measurement.

SYNTHETIC ILLUSTRATIVE - NOT A CVOEM RECORD